

Operating System

Project Report

Platform: Microsoft Azure

Operating System: Ubuntu Linux Configuration (Ubuntu1 & Ubuntu2)

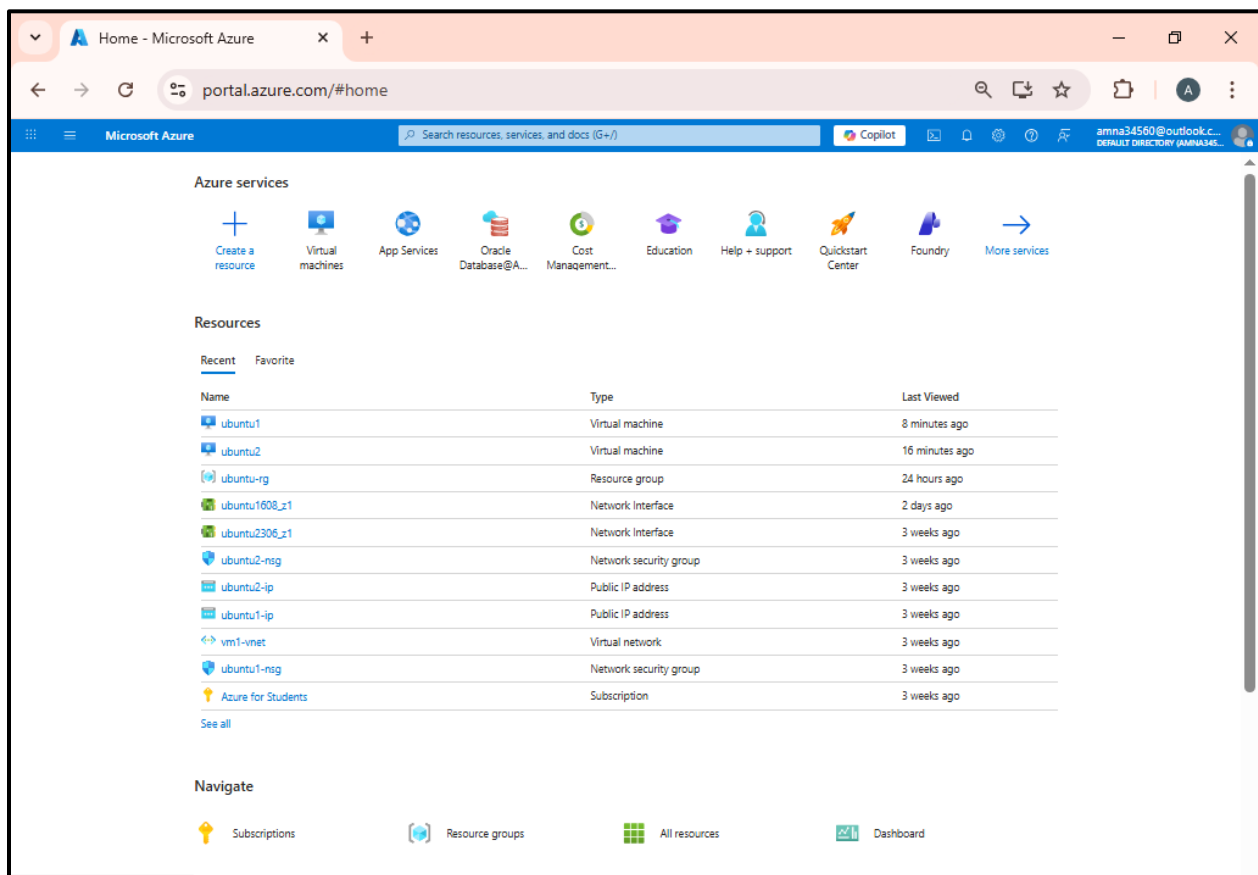
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1. Introduction

Cloud computing has transformed the way operating systems are deployed and managed. Instead of relying on physical hardware, organizations now use cloud platforms to run operating systems in virtualized environments. Microsoft Azure is one of the leading cloud platforms that allows users to create and manage virtual machines running various operating systems.

This project demonstrates the installation and configuration of Ubuntu Linux virtual machines on Microsoft Azure. The project further involves configuring multiple server services on Ubuntu Linux and verifying their functionality through client-server communication.



2. Objectives of the Project

The objectives of this project are:

- To deploy Ubuntu Linux virtual machines on Microsoft Azure
- To understand cloud-based operating system installation

- To configure and manage server services on Linux
- To establish secure communication using SSH
- To verify networking and service functionality through testing

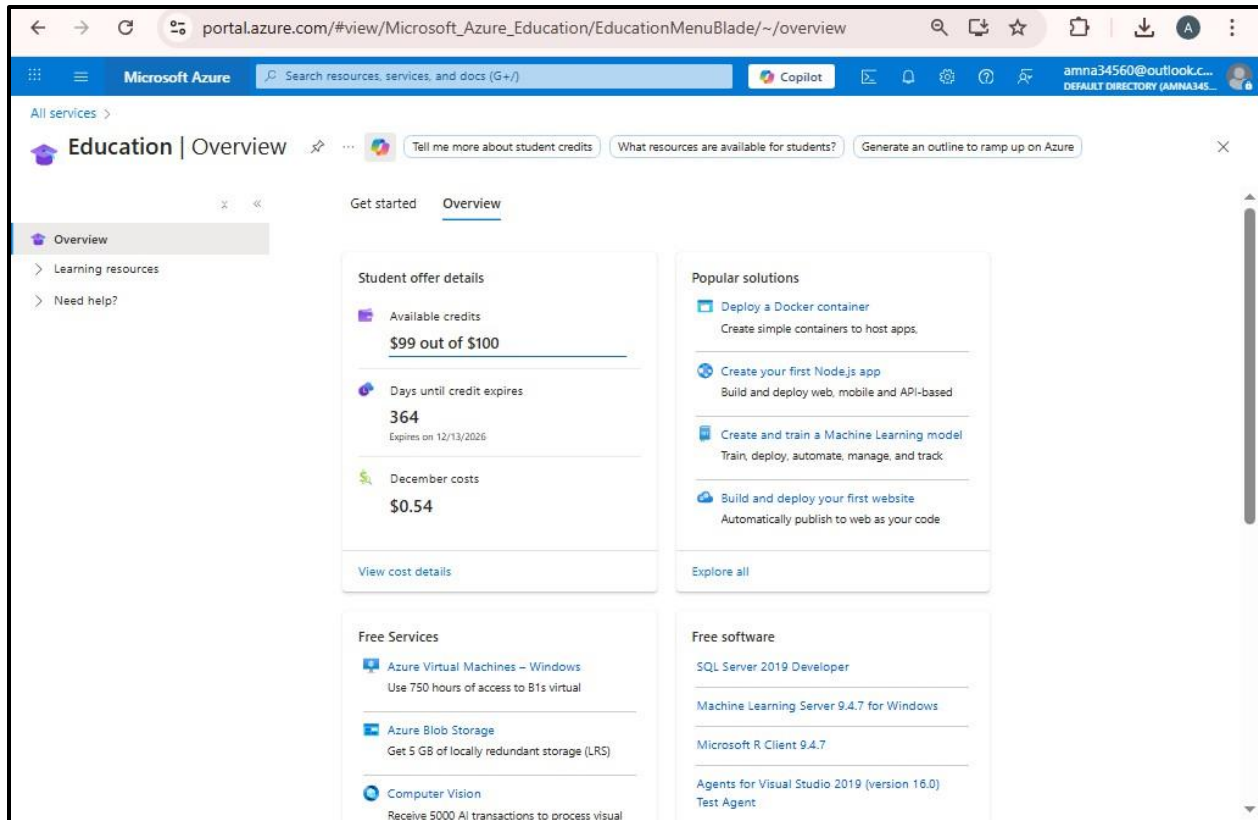
3. Tools and Technologies Used

Component	Description
Cloud Platform	Microsoft Azure
Operating System	Ubuntu Linux
Server Machine	Ubuntu1
Client Machine	Ubuntu2
Access Method	SSH
Server Services	FTP, Web, DNS, DHCP, Proxy

4. Phase 1: Virtual Machine Installation and Network Configuration

4.1 Azure Account Creation

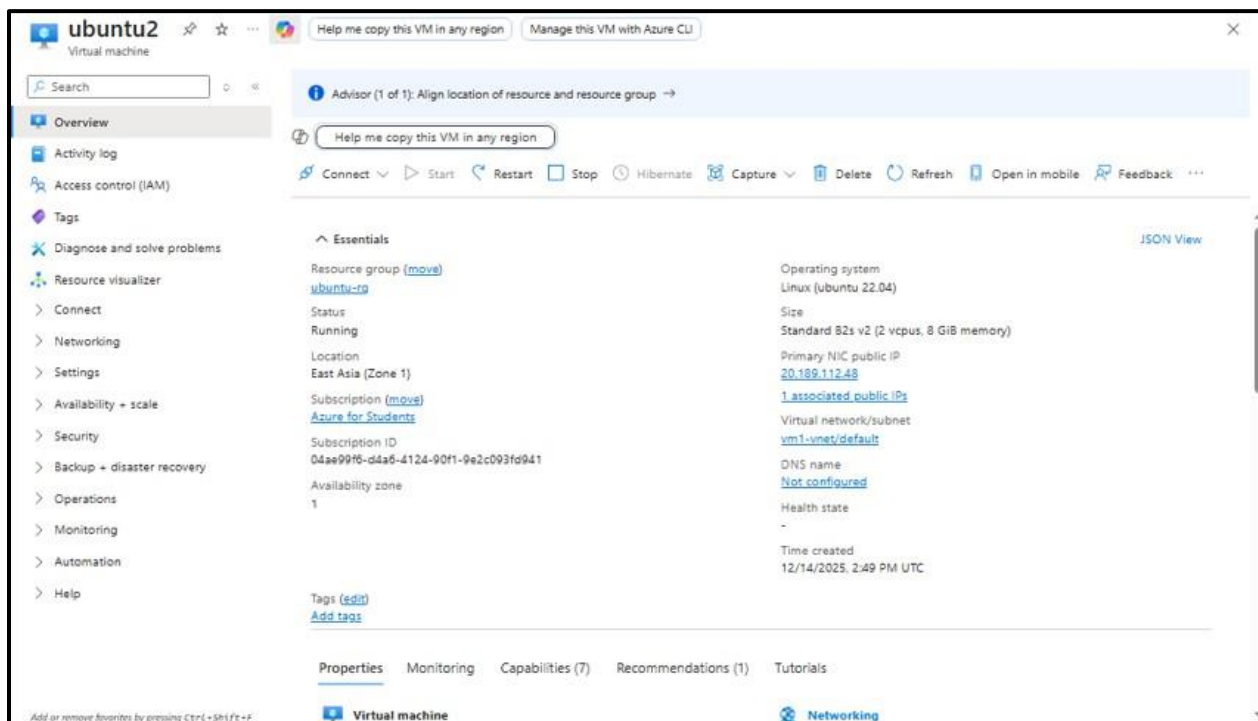
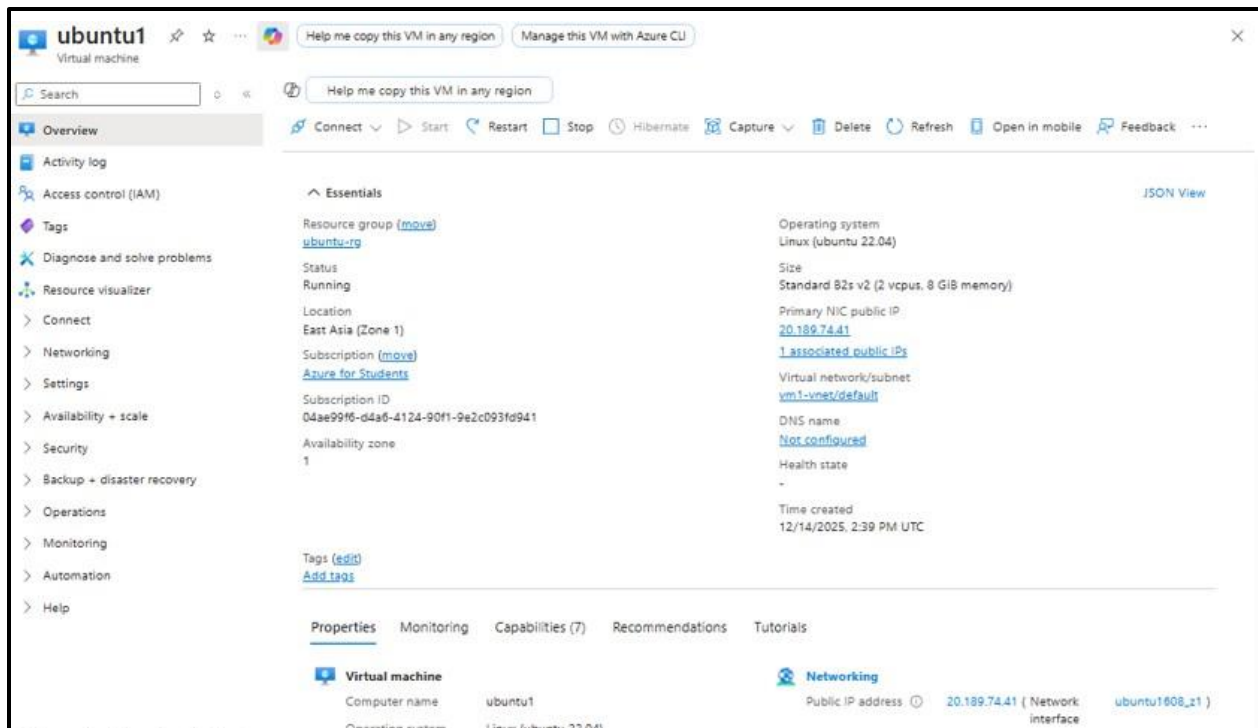
An Azure account was created using a Microsoft student email address. The **Azure for Students** subscription was activated to obtain free cloud credits for academic use.



4.2 Creation of Ubuntu Virtual Machines

Two Ubuntu Linux virtual machines were created in the Azure portal. Both virtual machines were placed in the same resource group and virtual network.

- **Ubuntu1:** Server Machine
- **Ubuntu2:** Client Machine



4.3 Secure Shell (SSH) Configuration

SSH was used to remotely access both Ubuntu virtual machines. Private key authentication was used for secure login from Windows PowerShell.

Example command:

ssh -i "E:\Download\keys\ubuntu1_key.pem" amnaa@20.189.74.41

```
amnaa@ubuntu1: ~  
Windows PowerShell  
Copyright (C) Microsoft Corporation. All rights reserved.  
  
Try the new cross-platform PowerShell https://aka.ms/pscore6  
  
PS C:\Users\scomputer> ssh -i "E:\Download\keys\ubuntu1_key.pem" amnaa@20.189.74.41  
>>  
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1044-azure x86_64)  
  
 * Documentation:  https://help.ubuntu.com  
 * Management:    https://landscape.canonical.com  
 * Support:       https://ubuntu.com/pro  
  
System information as of Sun Dec 14 15:15:13 UTC 2025  
  
System load:  0.0          Processes:            127  
Usage of /:   6.4% of 28.89GB Users logged in:      0  
Memory usage: 4%          IPv4 address for eth0: 10.0.0.4  
Swap usage:   0%  
  
Expanded Security Maintenance for Applications is not enabled.  
  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
amnaa@ubuntu1: $
```

```
ssh -i "E:\Download\keys\ubuntu2_key.pem" amnaa@20.189.112.48
```

```
amnaa@ubuntu2: ~  
Windows PowerShell  
Copyright (C) Microsoft Corporation. All rights reserved.  
  
Try the new cross-platform PowerShell https://aka.ms/pscore6  
  
PS C:\Users\scomputer> ssh -i "E:\Download\keys\ubuntu2_key.pem" amnaa@20.189.112.48  
>>  
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1044-azure x86_64)  
  
 * Documentation:  https://help.ubuntu.com  
 * Management:    https://landscape.canonical.com  
 * Support:       https://ubuntu.com/pro  
  
System information as of Sun Dec 14 15:58:23 UTC 2025  
  
System load:  0.08      Processes:            129  
Usage of /:   6.4% of 28.89GB Users logged in:      0  
Memory usage: 4%      IPv4 address for eth0: 10.0.0.5  
Swap usage:   0%  
  
Expanded Security Maintenance for Applications is not enabled.  
  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
amnaa@ubuntu2: $
```

4.4 Network Connectivity Testing

Network connectivity between Ubuntu1 and Ubuntu2 was tested using the ping command over private IP addresses.


```
amnaa@ubuntu1: $  
amnaa@ubuntu1: $ ping 10.0.0.5  
PING 10.0.0.5 (10.0.0.5) 56(84) bytes of data:  
64 bytes from 10.0.0.5: icmp_seq=1 ttl=64 time=1.52 ms  
64 bytes from 10.0.0.5: icmp_seq=2 ttl=64 time=1.09 ms  
64 bytes from 10.0.0.5: icmp_seq=3 ttl=64 time=0.855 ms  
64 bytes from 10.0.0.5: icmp_seq=4 ttl=64 time=0.809 ms  
64 bytes from 10.0.0.5: icmp_seq=5 ttl=64 time=0.772 ms  
64 bytes from 10.0.0.5: icmp_seq=6 ttl=64 time=0.784 ms  
64 bytes from 10.0.0.5: icmp_seq=7 ttl=64 time=0.823 ms  
64 bytes from 10.0.0.5: icmp_seq=8 ttl=64 time=0.758 ms  
64 bytes from 10.0.0.5: icmp_seq=9 ttl=64 time=1.01 ms  
64 bytes from 10.0.0.5: icmp_seq=10 ttl=64 time=6.79 ms  
64 bytes from 10.0.0.5: icmp_seq=11 ttl=64 time=0.677 ms  
64 bytes from 10.0.0.5: icmp_seq=12 ttl=64 time=0.739 ms  
64 bytes from 10.0.0.5: icmp_seq=13 ttl=64 time=0.670 ms  
64 bytes from 10.0.0.5: icmp_seq=14 ttl=64 time=0.842 ms  
64 bytes from 10.0.0.5: icmp_seq=15 ttl=64 time=0.822 ms  
64 bytes from 10.0.0.5: icmp_seq=16 ttl=64 time=0.738 ms  
64 bytes from 10.0.0.5: icmp_seq=17 ttl=64 time=6.42 ms  
64 bytes from 10.0.0.5: icmp_seq=18 ttl=64 time=0.762 ms  
64 bytes from 10.0.0.5: icmp_seq=19 ttl=64 time=0.838 ms  
64 bytes from 10.0.0.5: icmp_seq=20 ttl=64 time=1.67 ms  
64 bytes from 10.0.0.5: icmp_seq=21 ttl=64 time=0.641 ms  
64 bytes from 10.0.0.5: icmp_seq=22 ttl=64 time=3.40 ms  
64 bytes from 10.0.0.5: icmp_seq=23 ttl=64 time=0.698 ms  
64 bytes from 10.0.0.5: icmp_seq=24 ttl=64 time=0.763 ms  
64 bytes from 10.0.0.5: icmp_seq=25 ttl=64 time=0.795 ms  
64 bytes from 10.0.0.5: icmp_seq=26 ttl=64 time=0.793 ms  
64 bytes from 10.0.0.5: icmp_seq=27 ttl=64 time=0.813 ms  
64 bytes from 10.0.0.5: icmp_seq=28 ttl=64 time=0.958 ms  
64 bytes from 10.0.0.5: icmp_seq=29 ttl=64 time=0.846 ms  
64 bytes from 10.0.0.5: icmp_seq=30 ttl=64 time=0.829 ms  
64 bytes from 10.0.0.5: icmp_seq=31 ttl=64 time=3.68 ms  
64 bytes from 10.0.0.5: icmp_seq=32 ttl=64 time=0.879 ms  
64 bytes from 10.0.0.5: icmp_seq=33 ttl=64 time=1.31 ms  
64 bytes from 10.0.0.5: icmp_seq=34 ttl=64 time=2.80 ms  
64 bytes from 10.0.0.5: icmp_seq=35 ttl=64 time=0.691 ms  
64 bytes from 10.0.0.5: icmp_seq=36 ttl=64 time=0.814 ms  
64 bytes from 10.0.0.5: icmp_seq=37 ttl=64 time=0.812 ms  
64 bytes from 10.0.0.5: icmp_seq=38 ttl=64 time=11.9 ms  
64 bytes from 10.0.0.5: icmp_seq=39 ttl=64 time=0.807 ms  
64 bytes from 10.0.0.5: icmp_seq=40 ttl=64 time=0.720 ms  
64 bytes from 10.0.0.5: icmp_seq=41 ttl=64 time=12.4 ms  
64 bytes from 10.0.0.5: icmp_seq=42 ttl=64 time=0.777 ms  
64 bytes from 10.0.0.5: icmp_seq=43 ttl=64 time=8.98 ms  
64 bytes from 10.0.0.5: icmp_seq=44 ttl=64 time=0.972 ms
```

```
64 bytes from 10.0.0.5: icmp_seq=56 ttl=64 time=0.864 ms  
64 bytes from 10.0.0.5: icmp_seq=57 ttl=64 time=0.847 ms  
64 bytes from 10.0.0.5: icmp_seq=58 ttl=64 time=2.09 ms  
64 bytes from 10.0.0.5: icmp_seq=59 ttl=64 time=1.40 ms  
64 bytes from 10.0.0.5: icmp_seq=60 ttl=64 time=10.7 ms  
64 bytes from 10.0.0.5: icmp_seq=61 ttl=64 time=0.885 ms  
^C  
--- 10.0.0.5 ping statistics ---  
61 packets transmitted, 61 received, 0% packet loss, time 61050ms  
rtt min/avg/max/mdev = 0.555/1.873/12.398/2.698 ms  
amnaa@ubuntu1: $
```

```
amnaa@ubuntu2: ~  
Expanded Security Maintenance for Applications is not enabled.  
  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
New release '24.04.3 LTS' available.  
Run 'do-release-upgrade' to upgrade to it.  
  
Last login: Sun Dec 14 15:58:24 2025 from 154.192.18.4  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
amnaa@ubuntu2: $ ping 10.0.0.4  
PING 10.0.0.4 (10.0.0.4) 56(84) bytes of data:  
64 bytes from 10.0.0.4: icmp_seq=1 ttl=64 time=0.757 ms  
64 bytes from 10.0.0.4: icmp_seq=2 ttl=64 time=0.905 ms  
64 bytes from 10.0.0.4: icmp_seq=3 ttl=64 time=0.900 ms  
64 bytes from 10.0.0.4: icmp_seq=4 ttl=64 time=1.51 ms  
64 bytes from 10.0.0.4: icmp_seq=5 ttl=64 time=0.870 ms  
64 bytes from 10.0.0.4: icmp_seq=6 ttl=64 time=0.815 ms  
64 bytes from 10.0.0.4: icmp_seq=7 ttl=64 time=3.91 ms  
64 bytes from 10.0.0.4: icmp_seq=8 ttl=64 time=0.852 ms  
64 bytes from 10.0.0.4: icmp_seq=9 ttl=64 time=2.91 ms  
64 bytes from 10.0.0.4: icmp_seq=10 ttl=64 time=1.44 ms  
64 bytes from 10.0.0.4: icmp_seq=11 ttl=64 time=2.81 ms  
64 bytes from 10.0.0.4: icmp_seq=12 ttl=64 time=0.886 ms  
64 bytes from 10.0.0.4: icmp_seq=13 ttl=64 time=0.786 ms  
64 bytes from 10.0.0.4: icmp_seq=14 ttl=64 time=5.94 ms  
64 bytes from 10.0.0.4: icmp_seq=15 ttl=64 time=1.69 ms  
64 bytes from 10.0.0.4: icmp_seq=16 ttl=64 time=1.56 ms  
64 bytes from 10.0.0.4: icmp_seq=17 ttl=64 time=0.767 ms  
64 bytes from 10.0.0.4: icmp_seq=18 ttl=64 time=0.924 ms  
64 bytes from 10.0.0.4: icmp_seq=19 ttl=64 time=6.16 ms  
64 bytes from 10.0.0.4: icmp_seq=20 ttl=64 time=0.845 ms  
64 bytes from 10.0.0.4: icmp_seq=21 ttl=64 time=0.864 ms  
64 bytes from 10.0.0.4: icmp_seq=22 ttl=64 time=1.03 ms  
64 bytes from 10.0.0.4: icmp_seq=23 ttl=64 time=0.977 ms  
64 bytes from 10.0.0.4: icmp_seq=24 ttl=64 time=1.12 ms  
^C  
--- 10.0.0.4 ping statistics ---  
24 packets transmitted, 24 received, 0% packet loss, time 23363ms  
rtt min/avg/max/mdev = 0.757/1.717/6.163/1.522 ms  
amnaa@ubuntu2: $
```

Result: Successful ping responses confirmed proper internal network communication.

5. Phase 2: Server Installation and Configuration

5.1 FTP Server Configuration (vsftpd)

Purpose:

The FTP server allows file transfer between client and server machines.

Installation:

```
sudo apt update
sudo apt install vsftpd -y
```

Configuration:

The /etc/vsftpd.conf file was edited to enable local users and disable anonymous access.

```
amnaa@ubuntu1: $ sudo apt update
Hit:1 http://azure.archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://azure.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [3161 kB]
Get:6 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main Translation-en [484 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 c-n-f Metadata [19.0 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packages [5043 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 c-n-f Metadata [644 B]
Get:10 http://azure.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1244 kB]
Get:11 http://azure.archive.ubuntu.com/ubuntu jammy-updates/universe Translation-en [310 kB]
Get:12 http://azure.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 c-n-f Metadata [30.0 kB]
Get:13 http://azure.archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 Packages [57.6 kB]
Get:14 http://azure.archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 c-n-f Metadata [600 B]
Get:15 http://azure.archive.ubuntu.com/ubuntu jammy-backports/main amd64 Packages [69.4 kB]
Get:16 http://azure.archive.ubuntu.com/ubuntu jammy-backports/main amd64 c-n-f Metadata [412 B]
Get:17 http://azure.archive.ubuntu.com/ubuntu jammy-backports/universe amd64 Packages [31.7 kB]
Get:18 http://azure.archive.ubuntu.com/ubuntu jammy-backports/universe Translation-en [16.9 kB]
Get:19 http://azure.archive.ubuntu.com/ubuntu jammy-backports/universe amd64 c-n-f Metadata [672 B]
Get:20 http://azure.archive.ubuntu.com/ubuntu jammy-security/main amd64 Packages [2899 kB]
Get:21 http://azure.archive.ubuntu.com/ubuntu jammy-security/main Translation-en [417 kB]
Get:22 http://azure.archive.ubuntu.com/ubuntu jammy-security/main amd64 c-n-f Metadata [14.0 kB]
Get:23 http://azure.archive.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [4883 kB]
Get:24 http://azure.archive.ubuntu.com/ubuntu jammy-security/restricted Translation-en [917 kB]
Fetched 20.0 MB in 3s (7877 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
amnaa@ubuntu1: $
```



```
amnaa@ubuntu1: $ sudo apt install vsftpd -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ssl-cert
The following NEW packages will be installed:
  ssl-cert vsftpd
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 140 kB of archives.
After this operation, 391 kB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu jammy/main amd64 ssl-cert all 1.1.2 [17.4 kB]
Get:2 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 vsftpd amd64 3.0.5-0ubuntu1.1 [123 kB]
Fetched 140 kB in 0s (358 kB/s)
Preconfiguring packages ...
Selecting previously unselected package ssl-cert.
(Reading database ... 62847 files and directories currently installed.)
Preparing to unpack .../ssl-cert_1.1.2_all.deb ...
Unpacking ssl-cert (1.1.2) ...
Selecting previously unselected package vsftpd.
Preparing to unpack .../vsftpd_3.0.5-0ubuntu1.1_amd64.deb ...
Unpacking vsftpd (3.0.5-0ubuntu1.1) ...
Setting up ssl-cert (1.1.2) ...
Setting up vsftpd (3.0.5-0ubuntu1.1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/vsftpd.service → /lib/systemd/system/vsftpd.service.
Processing triggers for man-db (2.10.2-1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
```

```
amnaa@ubuntu1: $ sudo systemctl status vsftpd
● vsftpd.service - vsftpd FTP server
   Loaded: loaded (/lib/systemd/system/vsftpd.service; enabled; vendor preset: enabled)
   Active: active (running) since Fri 2026-01-02 12:41:04 UTC; 19s ago
     Process: 1548 ExecStartPre=/bin/mkdir -p /var/run/vsftpd/empty (code=exited, status=0/SUCCESS)
    Main PID: 1549 (vsftpd)
      Tasks: 1 (limit: 9522)
     Memory: 872.0K
        CPU: 4ms
    CGroup: /system.slice/vsftpd.service
            └─1549 /usr/sbin/vsftpd /etc/vsftpd.conf

Jan 02 12:41:04 ubuntu1 systemd[1]: Starting vsftpd FTP server...
Jan 02 12:41:04 ubuntu1 systemd[1]: Started vsftpd FTP server.
amnaa@ubuntu1: $
```

```
amnaa@ubuntu1: ~  
GNU nano 6.2 /etc/vsftpd.conf *  
#  
# Run standalone? vsftpd can run either from an inetd or as a standalone  
# daemon started from an initscript.  
listen=YES  
#  
# This directive enables listening on IPv6 sockets. By default, listening  
# on the IPv6 "any" address (:::) will accept connections from both IPv6  
# and IPv4 clients. It is not necessary to listen on *both* IPv4 and IPv6  
# sockets. If you want that (perhaps because you want to listen on specific  
# addresses) then you must run two copies of vsftpd with two configuration  
# files.  
listen_ipv6=NO  
#  
# Allow anonymous FTP? (Disabled by default).  
anonymous_enable=NO  
#  
# Uncomment this to allow local users to log in.  
  
local_enable=YES  
write_enable=YES  
#  
# Uncomment this to enable any form of FTP write command.  
#write_enable=YES  
#
```

Testing:

```
amnaa@ubuntu2:~$ ftp 10.0.0.4  
Connected to 10.0.0.4.  
220 (vsFTPd 3.0.5)  
Name (10.0.0.4:amnaa): amnaa  
331 Please specify the password.  
Password:  
230 Login successful.  
Remote system type is UNIX.  
Using binary mode to transfer files.  
ftp> pwd  
Remote directory: /  
ftp> bye  
221 Goodbye.  
amnaa@ubuntu2:~$
```

File upload successful

```
amnaa@ubuntu2: $ echo "FTP test file" > test.txt
amnaa@ubuntu2: $ ls
test.txt
amnaa@ubuntu2: $ ftp 10.0.0.4
Connected to 10.0.0.4.
220 (vsFTPd 3.0.5)
Name (10.0.0.4:amnaa): amnaa
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> put test.txt
local: test.txt remote: test.txt
229 Entering Extended Passive Mode (|||48047|)
150 Ok to send data.
100% |*****| 14 310.72 KiB/s 00:00 ETA
226 Transfer complete.
14 bytes sent in 00:00 (62.71 KiB/s)
ftp> ls
229 Entering Extended Passive Mode (|||38839|)
150 Here comes the directory listing.
-rw-rw-r-- 1 1000 1000 14 Jan 04 12:21 test.txt
226 Directory send OK.
```

5.2 Web Server Configuration (Apache)

Purpose:

Apache Web Server is used to host web pages using HTTP protocol.

Installation:

```
sudo apt install apache2 -y
```

Network Configuration:

An inbound rule allowing HTTP traffic on port 80 was added in the Azure Network Security Group.

```

amnaa@ubuntu1: $ sudo apt install apache2 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils bzip2 libapr1 libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap liblua5.3-0 mailcap mime-support
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser bzip2-doc
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils bzip2 libapr1 libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap liblua5.3-0 mailcap
mime-support
0 upgraded, 12 newly installed, 0 to remove and 0 not upgraded.
Need to get 2125 kB of archives.
After this operation, 8462 kB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libapr1 amd64 1.7.0-8ubuntu0.22.04.2 [108 kB]
Get:2 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1 amd64 1.6.1-5ubuntu4.22.04.2 [92.8 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.1-5ubuntu4.22.04.2 [11.3 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1-ldap amd64 1.6.1-5ubuntu4.22.04.2 [9170 B]
Get:5 http://azure.archive.ubuntu.com/ubuntu jammy/main amd64 liblua5.3-0 amd64 5.3.6-1build1 [140 kB]
Get:6 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-bin amd64 2.4.52-1ubuntu4.16 [1349 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-data all 2.4.52-1ubuntu4.16 [165 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-utils amd64 2.4.52-1ubuntu4.16 [89.3 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu jammy/main amd64 mailcap all 3.70+nmu1ubuntu1 [23.8 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu jammy/main amd64 mime-support all 3.66 [3696 B]
Get:11 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2 amd64 2.4.52-1ubuntu4.16 [97.9 kB]
Get:12 http://azure.archive.ubuntu.com/ubuntu jammy/main amd64 bzip2 amd64 1.0.8-5build1 [34.8 kB]
Fetched 2125 kB in 1s (3223 kB/s)
Selecting previously unselected package libapr1:amd64.
(Reading database ... 62913 files and directories currently installed.)
Preparing to unpack .../00-libapr1_1.7.0-8ubuntu0.22.04.2_amd64.deb ...
Unpacking libapr1:amd64 (1.7.0-8ubuntu0.22.04.2) ...
Selecting previously unselected package libaprutil1:amd64.
Preparing to unpack .../01-libaprutil1_1.6.1-5ubuntu4.22.04.2_amd64.deb ...
Unpacking libaprutil1:amd64 (1.6.1-5ubuntu4.22.04.2) ...
Selecting previously unselected package libaprutil1-dbd-sqlite3:amd64.
Preparing to unpack .../02-libaprutil1-dbd-sqlite3_1.6.1-5ubuntu4.22.04.2_amd64.deb ...
Unpacking libaprutil1-dbd-sqlite3:amd64 (1.6.1-5ubuntu4.22.04.2) ...
Selecting previously unselected package libaprutil1-ldap:amd64.
Unpacking libaprutil1-ldap:amd64 (1.6.1-5ubuntu4.22.04.2) ...
Selecting previously unselected package liblua5.3-0:amd64.
Preparing to unpack .../04-liblua5.3-0_5.3.6-1build1_amd64.deb ...
Unpacking liblua5.3-0:amd64 (5.3.6-1build1) ...
Selecting previously unselected package apache2-bin.
Preparing to unpack .../05-apache2-bin_2.4.52-1ubuntu4.16_amd64.deb ...
Unpacking apache2-bin (2.4.52-1ubuntu4.16) ...
Selecting previously unselected package apache2-data.
Preparing to unpack .../06-apache2-data_2.4.52-1ubuntu4.16_all.deb ...
Unpacking apache2-data (2.4.52-1ubuntu4.16) ...
Selecting previously unselected package apache2-utils.

```

Microsoft Azure Search resources, services, and docs (G+)

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Network interface / IP configuration

ubuntu1608_z1 (primary) / ipconfig1 (primary)

Essentials

Network interface : ubuntu1608_z1

Virtual network / subnet : vm1-vnet / default

Public IP address : 20.189.74.41

Private IP address : 10.0.0.4

Admin security rules : 0 (Configure)

Rules

Network security group ubuntu1-nsg (attached to networkinterface: ubuntu1608_z1)

Impacts 0 subnets, 1 network interfaces

Search rules

Source == all Destination == all Protocol == all Action ==

Priority	Name	Port
300	SSH	22
65000	AllowVNetInBound	Any
65001	AllowAzureLoadBalancerInBound	Any
65500	DenyAllInBound	Any

Outbound port rules (3)

Add inbound security rule

Source: Any

Source port ranges: *

Destination: Any

Service: HTTP

Destination port ranges: 80

Protocol: TCP

Action: Allow

Priority: 310

Name: Allow_HTTP

Description:

Add Cancel Give feedback

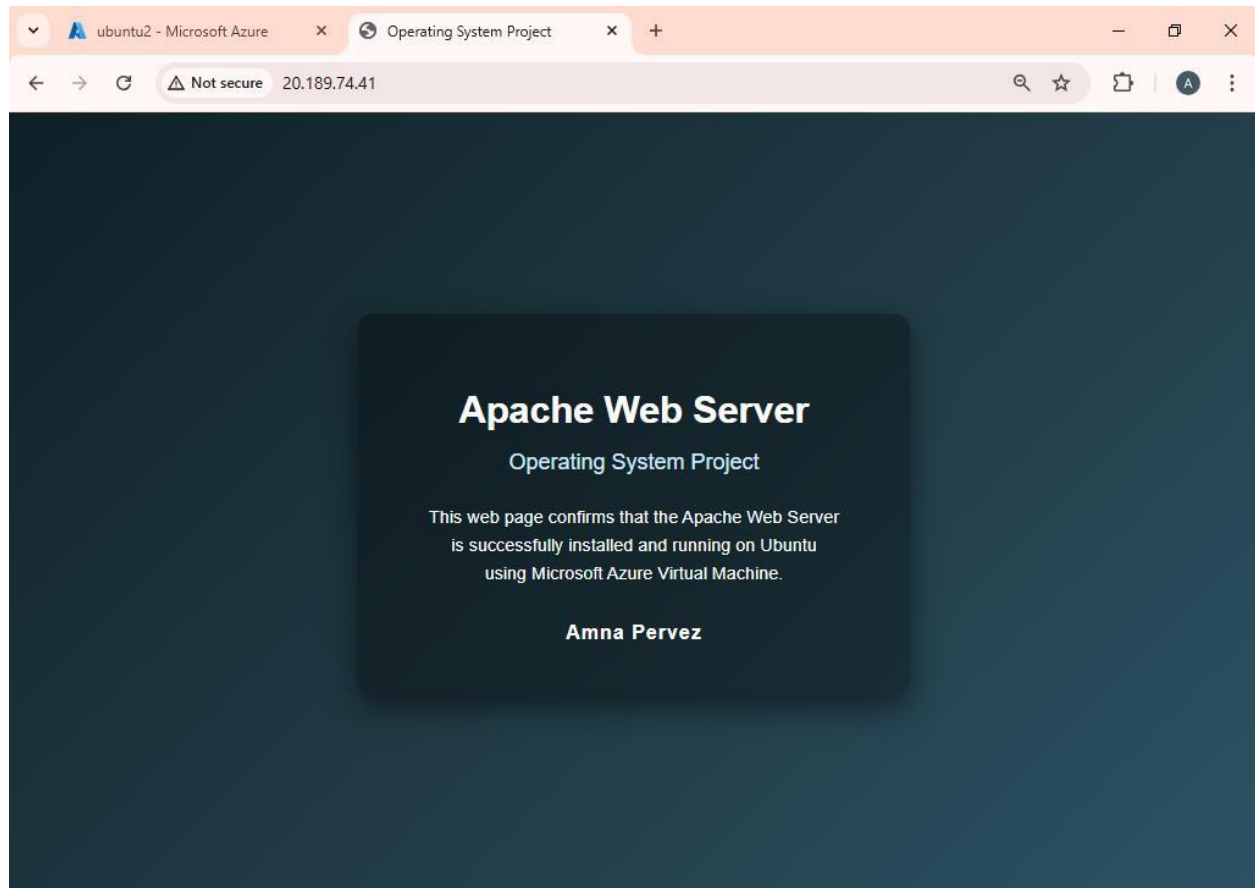
amnaa@ubuntu1: ~

GNU nano 6.2

/var/www/html/index.html

```
.container {
  background: rgba(0, 0, 0, 0.35);
  padding: 45px 65px;
  border-radius: 14px;
  text-align: center;
  box-shadow: 0 12px 30px rgba(0,0,0,0.5);
}
h1 {
  font-size: 34px;
  margin-bottom: 8px;
}
h2 {
  font-size: 20px;
  font-weight: normal;
  color: #cfe9ff;
  margin-bottom: 25px;
}
p {
  font-size: 16px;
  line-height: 1.6;
}
.name {
  margin-top: 30px;
  font-size: 18px;
  font-weight: bold;
  letter-spacing: 1px;
}
</style>
</head>
<body>
  <div class="container">
    <h1>Apache Web Server</h1>
    <h2>Operating System Project</h2>
    <p>
      This web page confirms that the Apache Web Server<br>
      is successfully installed and running on Ubuntu<br>
      using Microsoft Azure Virtual Machine.
    </p>
    <div class="name">
      Amna Pervez
    </div>
  </div>
</body>
</html>
```

Testing:



5.3 DNS Server Configuration (Bind9)

Purpose:

DNS resolves domain names into IP addresses.

Installation:

```
sudo apt install bind9 -y
```

```
Jan 02 12:52:01 ubuntu1 systemd[1]: Starting The Apache HTTP Server...
Selecting previously unselected package bind9-utils.
(Reading database ... 63673 files and directories currently installed.)
Preparing to unpack .../bind9-utils_1%3a9.18.39-0ubuntu0.22.04.2_amd64.deb ...
Unpacking bind9-utils (1:9.18.39-0ubuntu0.22.04.2) ...
Selecting previously unselected package dns-root-data.
Preparing to unpack .../dns-root-data_2024071801-ubuntu0.22.04.1_all.deb ...
Unpacking dns-root-data (2024071801-ubuntu0.22.04.1) ...
Selecting previously unselected package bind9.
Preparing to unpack .../bind9_1%3a9.18.39-0ubuntu0.22.04.2_amd64.deb ...
Unpacking bind9 (1:9.18.39-0ubuntu0.22.04.2) ...
Setting up dns-root-data (2024071801-ubuntu0.22.04.1) ...
Setting up bind9-utils (1:9.18.39-0ubuntu0.22.04.2) ...
Setting up bind9 (1:9.18.39-0ubuntu0.22.04.2) ...
Adding group `bind' (GID 125) ...
Done.
Adding system user `bind' (UID 115) ...
Adding new user `bind' (UID 115) with group `bind' ...
Not creating home directory `/var/cache/bind'.
wrote key file "/etc/bind/rndc.key"
named-resolvconf.service is a disabled or a static unit, not starting it.
Created symlink /etc/systemd/system/bind9.service → /lib/systemd/system/named.service.
Created symlink /etc/systemd/system/multi-user.target.wants/named.service → /lib/systemd/system/named.service.
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for ufw (0.36.1-4ubuntu0.1) ...
Scanning processes...
```

Testing:

nslookup google.com 10.0.0.4

```
amnaa@ubuntu2:~$ nslookup google.com 10.0.0.4
Server:      10.0.0.4
Address:     10.0.0.4#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.197.142
Name:   google.com
Address: 2404:6800:4005:822::200e
```

5.4 DHCP Server Configuration

Purpose:

DHCP automatically assigns IP addresses to clients.

Installation:

sudo apt install isc-dhcp-server -y

```

amnaa@ubuntu1: ~
amnaa@ubuntu1: $ sudo apt install isc-dhcp-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libirs-export161 libiscfg-export163
Suggested packages:
  isc-dhcp-server-ldap policycoreutils
The following NEW packages will be installed:
  isc-dhcp-server libirs-export161 libiscfg-export163
0 upgraded, 3 newly installed, 0 to remove and 0 not upgraded.
Need to get 529 kB of archives.
After this operation, 1546 kB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu jammy/main amd64 libiscfg-export163 amd64 1:9.11.19+dfsg-2.1ubuntu3 [53.0 kB]
Get:2 http://azure.archive.ubuntu.com/ubuntu jammy/main amd64 libirs-export161 amd64 1:9.11.19+dfsg-2.1ubuntu3 [20.0 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 isc-dhcp-server amd64 4.4.1-2.3ubuntu2.4 [456 kB]
Fetched 529 kB in 1s (920 kB/s)
Preconfiguring packages ...
Selecting previously unselected package libiscfg-export163.
(Reading database ... 63771 files and directories currently installed.)
Preparing to unpack .../libiscfg-export163_1%3a9.11.19+dfsg-2.1ubuntu3_amd64.deb ...
Unpacking libiscfg-export163 (1:9.11.19+dfsg-2.1ubuntu3) ...
Selecting previously unselected package libirs-export161.
Preparing to unpack .../libirs-export161_1%3a9.11.19+dfsg-2.1ubuntu3_amd64.deb ...
Unpacking libirs-export161 (1:9.11.19+dfsg-2.1ubuntu3) ...
Selecting previously unselected package isc-dhcp-server.
Preparing to unpack .../isc-dhcp-server_4.4.1-2.3ubuntu2.4_amd64.deb ...
Unpacking isc-dhcp-server (4.4.1-2.3ubuntu2.4) ...
Setting up libiscfg-export163 (1:9.11.19+dfsg-2.1ubuntu3) ...
Setting up libirs-export161 (1:9.11.19+dfsg-2.1ubuntu3) ...
Setting up isc-dhcp-server (4.4.1-2.3ubuntu2.4) ...
Generating /etc/default/isc-dhcp-server...
Created symlink /etc/systemd/system/multi-user.target.wants/isc-dhcp-server.service → /lib/systemd/system/isc-dhcp-server.service.
Created symlink /etc/systemd/system/multi-user.target.wants/isc-dhcp-server6.service → /lib/systemd/system/isc-dhcp-server6.service.
Processing triggers for libc-bin (2.35-0ubuntu3.11) ...
Processing triggers for man-db (2.10.2-1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.

```

```

amnaa@ubuntu1: ~
No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
amnaa@ubuntu1: $ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:0d:3a:81:fd:3b brd ff:ff:ff:ff:ff:ff
    inet 10.0.0.4/24 metric 100 brd 10.0.0.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::20d:3aff:fe81:fd3b/64 scope link
        valid_lft forever preferred_lft forever
3: enP10695s1: <BROADCAST,MULTICAST,SLAVE,UP,LOWER_UP> mtu 1500 qdisc mq master eth0 state UP group default qlen 1000
    link/ether 00:0d:3a:81:fd:3b brd ff:ff:ff:ff:ff:ff
    altname enP10695p0s2
    inet6 fe80::20d:3aff:fe81:fd3b/64 scope link
        valid_lft forever preferred_lft forever
amnaa@ubuntu1: $ sudo nano /etc/default/isc-dhcp-server
amnaa@ubuntu1: $

```

```
#}

subnet 10.0.0.0 netmask 255.255.255.0 {
    range 10.0.0.50 10.0.0.100;
    option routers 10.0.0.1;
    option domain-name-servers 10.0.0.4;
}

■
```

Testing:

ip a

```
amnaa@ubuntu2: $ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:0d:3a:80:45:79 brd ff:ff:ff:ff:ff:ff
    inet 10.0.0.5/24 metric 100 brd 10.0.0.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::20d:3aff:fe80:4579/64 scope link
        valid_lft forever preferred_lft forever
3: enP18747s1: <BROADCAST,MULTICAST,SLAVE,UP,LOWER_UP> mtu 1500 qdisc mq master eth0 state UP group default qlen 1000
    link/ether 00:0d:3a:80:45:79 brd ff:ff:ff:ff:ff:ff
    altname enP18747p0s2
    inet6 fe80::20d:3aff:fe80:4579/64 scope link
        valid_lft forever preferred_lft forever
amnaa@ubuntu2: $
```

5.5 Proxy Server Configuration (Squid)

Purpose:

A proxy server controls and monitors internet access.

Installation:

sudo apt install squid -y

```
Jan 02 13:56:28 ubuntu1 squid[5126]: Using Least Load store dir selection
Jan 02 13:56:28 ubuntu1 squid[5126]: Set Current Directory to /var/spool/squid
Jan 02 13:56:28 ubuntu1 squid[5126]: Finished loading MIME types and icons.
Jan 02 13:56:28 ubuntu1 squid[5126]: HTCP Disabled.
Jan 02 13:56:28 ubuntu1 squid[5126]: Pinger socket opened on FD 14
Jan 02 13:56:28 ubuntu1 squid[5126]: Squid plugin modules loaded: 0
Jan 02 13:56:28 ubuntu1 squid[5126]: Adaptation support is off.
Jan 02 13:56:28 ubuntu1 squid[5126]: Accepting HTTP Socket connections at conn3 local=[::]:3128 remote=[::] FD 12 flags=9
Jan 02 13:56:28 ubuntu1 systemd[1]: Started Squid Web Proxy Server.
Jan 02 13:56:29 ubuntu1 squid[5126]: storeLateRelease: released 0 objects
amnaa@ubuntu1: $ ■
```

```
amnaa@ubuntu1: ~  
GNU nano 6.2 /etc/squid/squid.conf *  
http_port 3128  
  
acl localnet src 10.0.0.0/24  
http_access allow localnet  
http_access deny all  
  
dns_v4_first on  
visible_hostname squidproxy
```

Testing:

```
amnaa@ubuntu2: $ curl -x http://10.0.0.4:3128 http://example.com  
<!doctype html><html lang="en"><head><title>Example Domain</title><meta name="viewport" content="width=device-width, initial-scale=1"><style>bo  
dy{background:#eee;width:60vw;margin:15vh auto;font-family:system-ui,sans-serif}h1{font-size:1.5em}div{opacity:0.8}a:link,a:visited{color:#348}  
</style><body><div><h1>Example Domain</h1><p>This domain is for use in documentation examples without needing permission. Avoid use in operatio  
ns.<p><a href="https://iana.org/domains/example">Learn more</a></div></body></html>  
amnaa@ubuntu2: $
```

6. Results and Discussion

All configured services were tested successfully from the client machine. The results confirm that Ubuntu Linux can effectively host multiple server services within a cloud environment when proper networking and security configurations are applied.

7. Conclusion

This project successfully demonstrated the deployment of Ubuntu Linux virtual machines on Microsoft Azure and the configuration of essential server services. The practical implementation improved understanding of operating systems, networking, and cloud-based infrastructure.

8. Future Scope

- Implement firewall security using UFW
- Enable HTTPS using SSL certificates
- Monitor services using system logs
- Deploy load-balanced servers